

TEAL Electronics Corporation

Model PET/MR MDP Panel

TEAL Part Number 3600005-03



1030370_04.doc

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SPECIFICATIONS

COMMON SPECIFICATIONS	ALL MODEL NUMBERS
<i>Rated Operational ambient</i>	0° to 40°C
INDIVIDUAL SPECIFICATIONS	MODEL NUMBERS
<i>Model Number</i>	PET/MR MDP PANEL
<i>(TEAL) Part Number</i>	3600005-03
<i>Overload Protection</i>	200 Amps
<i>Input Voltage Range</i>	380/400/415/480VAC +/- 10%
<i>Input Current</i>	150.4/142.9/137.7/119.1 Amps
<i>Output Voltages</i>	Output voltages is respective to input voltage
<i>Output Currents</i>	211 Amps Max at 380VAC
<i>Weight Pounds (Kg.)</i>	160 (72.7)
<i>RoHS Compliant</i>	Yes



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1- SAFETY



High voltage. Turn power off before servicing.



Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.



To reduce the risk of electric shock, install in a temperature and humidity controlled indoor area relatively free of conductive contaminants.

2 - PRODUCT DESCRIPTION

2-1 Product

The Main Disconnect Panel (MDP) cabinet is designed to be used in the PET/MR system to provide complete power distribution and Emergency-Off control for the Heat Exchanger Cabinet and the Power Distribution Unit (PDU). Power is provided directly from the facility through the Main Disconnect Panel to the entire PET/MR product.

2-2 Additional Features

The MDP will provide two Remote Emergency-Off Switches with Emergency-Off labels for the customer to mount at accessible locations in the Magnet Room and Equipment Room. Emergency-Off switches are normally closed push buttons. The MDP will also provide two jumpers wire 58 and 59 and installed inside the MDP to allow it to be operated prior to the installation of the customer's remote Emergency-Off switches.



Jumper wires 58 and 59 must be removed immediately upon the installation of the remote Emergency-Off switches to enable their operation.



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3- PHYSICAL INSTALLATION

3-1 Unpacking

- Protect finish, light, extended handles and buttons from damage while uncrating and placement.
- If forklift is used, lift only by frame, avoiding damage to sheet metal covers. If there is a removable door remove it while uncrating and during placement to avoid damage to hinges and latches.

If there is external damage or if internal components “rattle”, do not apply voltage. Check for any loose components or connections, call TEAL Tech Support for assistance.

All units are carefully and adequately packed. Please retain packing materials until the unit is satisfactorily install and accepted.

Damage in transit is always possible. Visible external damage should be noted on the bill of lading upon receipt and a claim filed with the shipper. Concealed damage can also be claimed with the shipper within 15 days. Contact TEAL if you wish assistance in filing your claim.

Each unit is inspected and tested before it leaves the factory. An individual check-out sheet is maintained on file by TEAL and retrievable by power size and serial number. Items tested include voltages input/output, current, and spike attenuation both normal and common mode. All connections are tight when they leave the factory.

3-2 Mounting

Refer to the attached mechanical drawing for dimensional information. The unit should be mounted in a dry location. The recommended upper limit installation height is 1.7m.

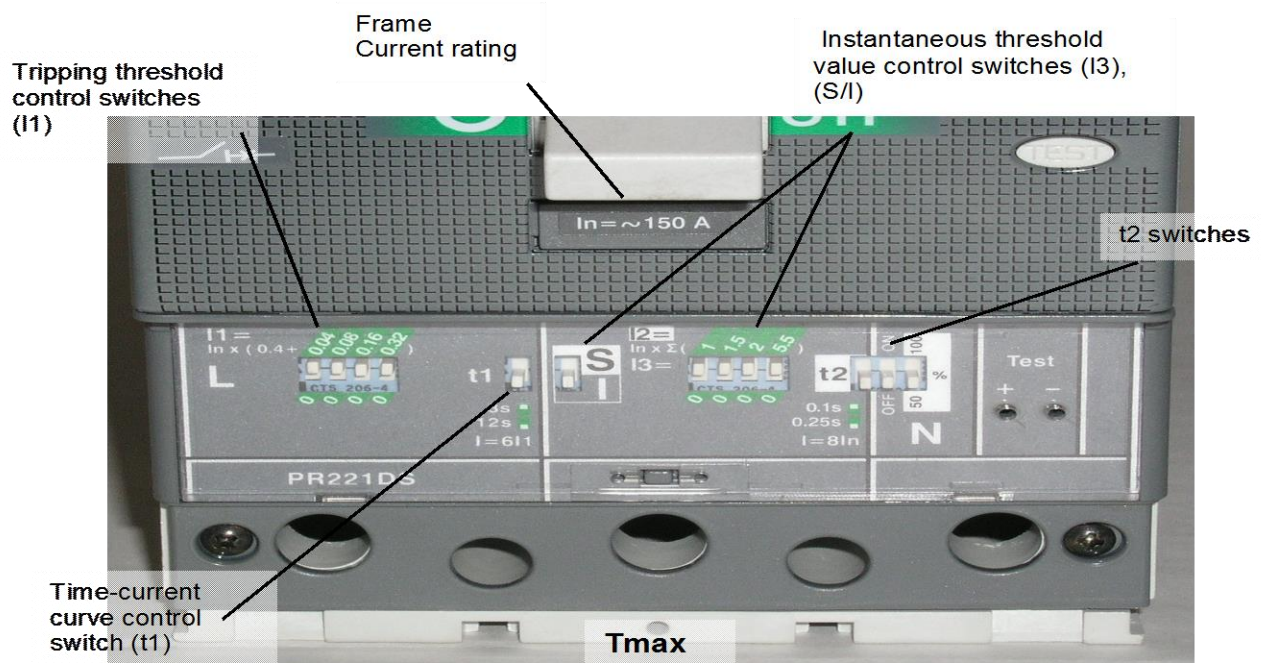


HIGH VOLTAGE. Turn power off before removing cover.

4- CONNECTIONS

Main Input connection will be to top of CB1. PDU Output will be connected to bottom of CB2 and HEC Output will be connected to top of CON1. Call TEAL Tech Support if you need advice on these connections.

4-1 ABB Option



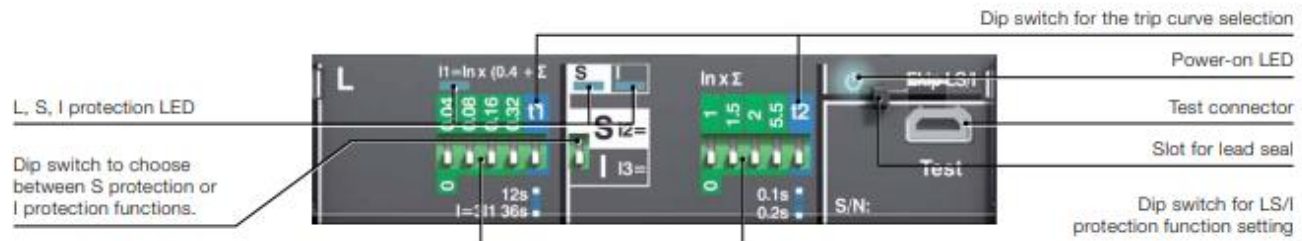
INPUT CB1 SETTINGS				
L SETTING	t1 SETTING	S/I SETTING	I3 SETTING	t2 SETTING
- -	-	-	- -	...

ABB T4 CIRCUIT BREAKER DIP SWITCH SETTINGS

Illustration 4-1



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INPUT CB1 SETTINGS				
L SETTING	t1 SETTING	S/I SETTING	I3 SETTING	t2 SETTING
0.14 0.18 0.15 0.32	3s 24s	5 1	1.0 1.5 2.0 5.0	0.1s 0.2s

ABB XT4 CIRCUIT BREAKER DIP SWITCH SETTINGS
Illustration 4-2

Three dip-switches on the circuit breaker panel control the input voltage overload setting, the overload delay, and the short-circuit trip adjustment. Factory settings for the three switches are:

- I1 - set based on the input voltage. Refer to the Schematic for a table of the other I1 settings.
- t1 - longest delay
- I/S – longest delay (set to I position)
- t2 – All dip switches in down position
- I3 – All four dip switches in up position

Use a small blade screwdriver to move the dip switch toggles to the appropriate settings.

5- MAINTENANCE

It is recommended that the MDP battery be replaced every two (2) years. Annually, the unit should be serviced by a qualified electrician to tighten all electrical connections. Any accumulations of dust or debris should be removed to keep ventilation working for convection cooling.

5-1 FRU List

Circuit diagrams, component part lists, descriptions, calibration instructions, or other information is available upon request from Teal Electronics for the purpose of repair or replacement. Refer to contact information in section 7.



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Ref. Designator	Description	Part number
AUX2	CONTACTOR, AUX CONTACT, 1N.O./N.C. ABB #CAL5-11	4990929
BATT	BATTERY, RECHARGEABLE LEAD ACID, 12V, 7AH POWERSONIC #PS-1270	4990906
BVUPS	RELAY, BATTERY VOLTAGE UPS, 12V, TRANSTRONICS #BVUPS12FPA-10	4990908
CB1	CB, 250.0A, 3P, 600V ABB #XT4NU3250EFF000XXX	4990930
CB2	CB, 150.0A, 3P, 600V, ABB #XT4NU3150AFF000XXX	4990992-01
CB3	CB, 50.0A, 3P, 480V ABB #XT1NU3050AAA000XXX	4990932
CB4	CB, 3.0A, 1P, 277/480YABB#S201-K3	4990933
CB5	CB, 2A, 1P, 500 VDC ABB #S281UC-K2	4990934
CB6	CB, 30.0A, 3P, 480V ABB #XT1NU3030AAA400XXX, W/ 110-125VAC/DC UV	4991156
K3	CONTACTOR, 65.0A, 120VAC 3P, ABB#AF65-30-11-13	4990935
F1-F2	CB, FUSE, 1.5A, 600VAC 13/32" X 1 1/2" BUSSMAN #FNQ-R-1 1/2	4990936
F3-F5	CB, FUSE, 60.0A, 600VAC/ 500VDC, 100Ka IR SHAWMUT #AJT60	4990937
F6-F7	CB, FUSE, 1.0A, 500V AC/DC FAST ACTING BUSSMAN #KLM-1	4990909
IND1	LIGHT, GREEN UNIT CASE, 120VAC ABB#CBK-KLFL1G	4990938
PS1	POWER SUPPLY, 15W, 12VDC OUTPUT, 100-240VAC INPUT PULS #ML15.121	4990907
RY1-RY2	RELAY, 10A, 12VDC, DPDT IDEC #RR2BA- UDC12V	4990939-01
SW1	SWITCH, N.C. BLOCK FOR CH PB's CUTLER HAMMER #E22B1	4990940
SW2	SWITCH, N.O. BLOCK FOR CH PB's CUTLER HAMMER #E22B2	4990941
T1	XFORMER, CNTRL, 300VA 208/240/380/416/480\120VSQD #9070T300D40	4990942

5-2 FRU Replacement Procedures

⚠ DANGER



High voltage. Turn power off before servicing.

TEAL

We Power Technology

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Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.



Schematics should be used when replacing parts for wiring purposes.

- **Battery, PowerSonic # PS-1270, TEAL P/N 4990906**
 - Reference Designator: BATT**
 - Tools: Philips head screwdriver**
 - Size of Mounting Hardware: 10x32 Screws**
- 1. Open front door on panel
- 2. Disconnect wires 50 & 55 from the battery
- 3. Remove the 4 screws securing the battery cover to the component plate





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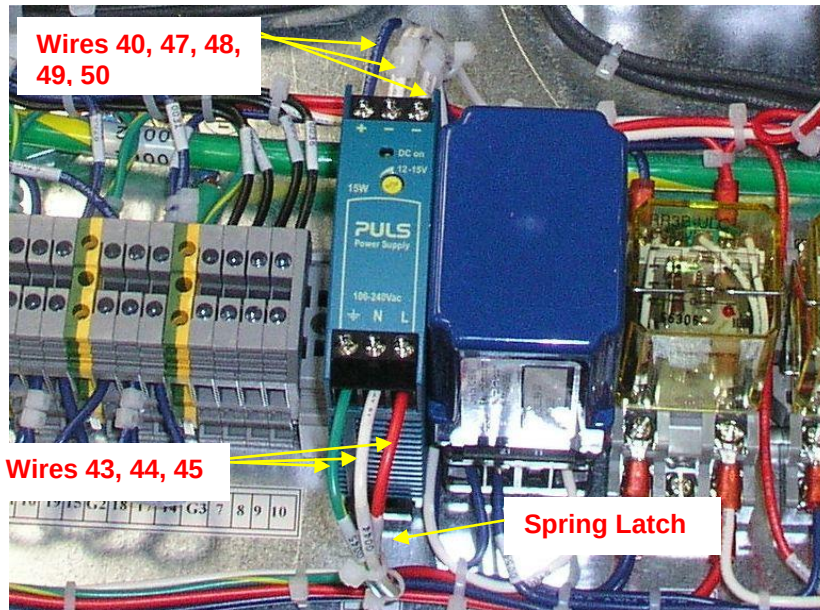
4. Remove the old battery
5. Installation is the reverse of removal. It is important to make sure the positive wire (55) goes to the + terminal on the battery

- **Power Supply, PULSPower # ML15.121, TEAL P/N 4990907**

Reference Designator: PS1

Tools: Philips head screwdriver, Flat head screwdriver

1. Open front door on panel



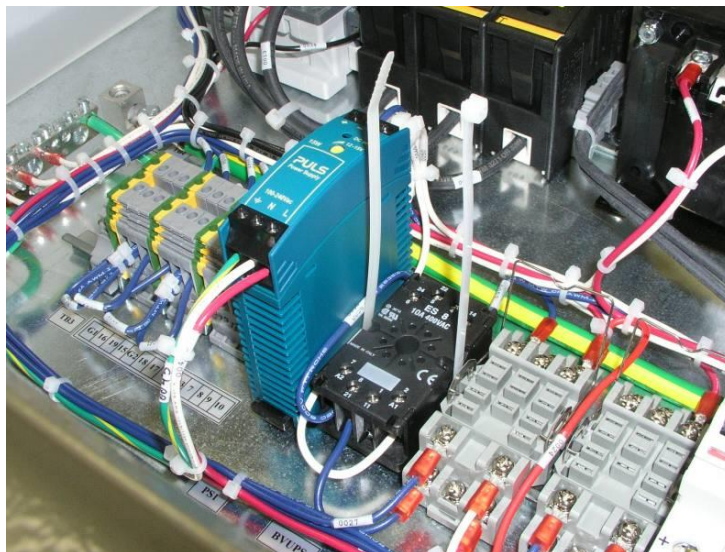
2. Disconnect wires 40, 43, 44, 45, 47, 48, 49 & 50 from the power supply
3. Use a flat head screw driver to release the spring clip holding the power supply to the DIN rail
4. Installation is the reverse of removal

- **UPS, Transtronics # BVUPS12PFA-10, TEAL P/N 4990908**

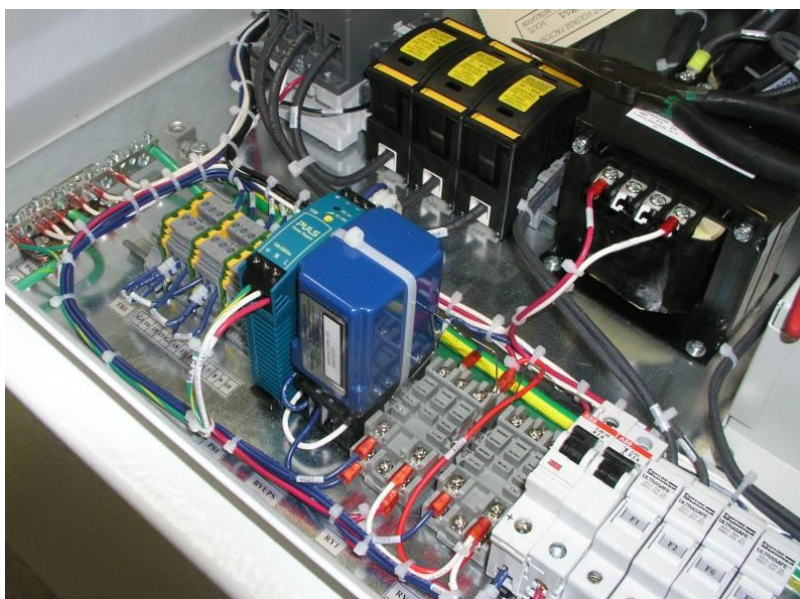
Reference Designator: BVUPS

Tools: Wire snips

1. Open front door on panel
2. Cut the tie-wraps securing the UPS to the DIN rail
3. Remove the old UPS
4. Remove the UPS base from the DIN rail by pushing to one side and lifting off the rail
5. Install new tie-wraps onto the base and reinstall onto the DIN rail



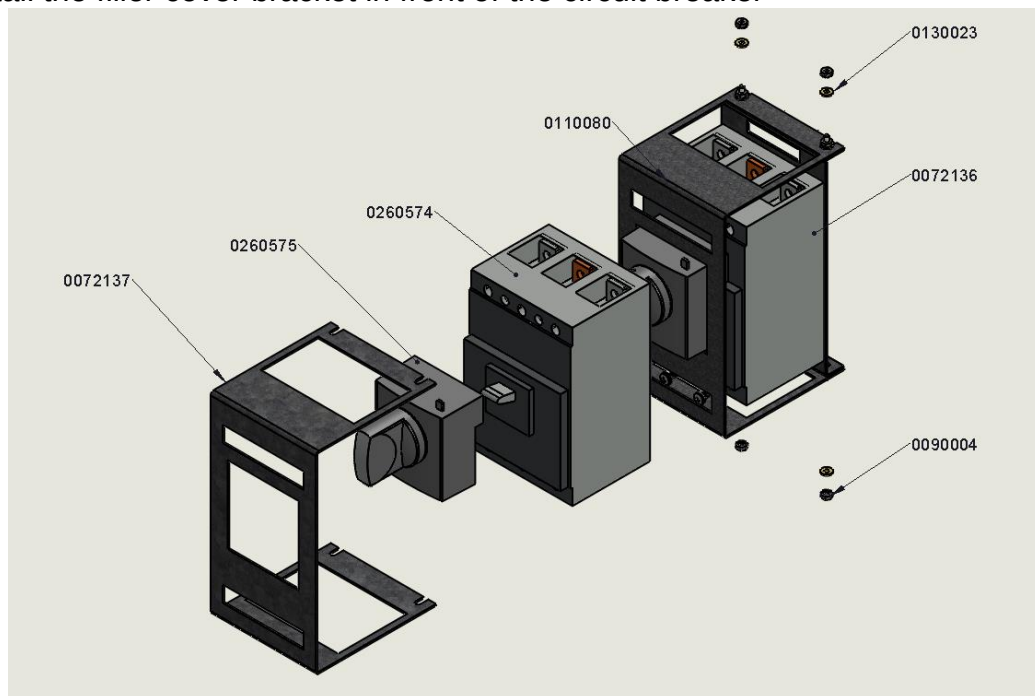
6. Install new UPS and secure using tie-wrap



- **Circuit Breaker, ABB # XT4NU3250EFF000XXX, TEAL P/N 4990930**
Reference Designator: CB1
Tools: Phillips head screwdriver
 1. Open front door on panel
 2. Disconnect wires 1, 2, and 3 from the circuit breaker



3. Using a Phillips head screwdriver, remove the screws holding the circuit breaker to the plate
4. Remove the old circuit breaker
5. Install the elevation bracket
6. Using the provided hardware, install the new circuit breaker
7. Install the filler cover bracket in front of the circuit breaker



8. Connect the wires removed in step 2



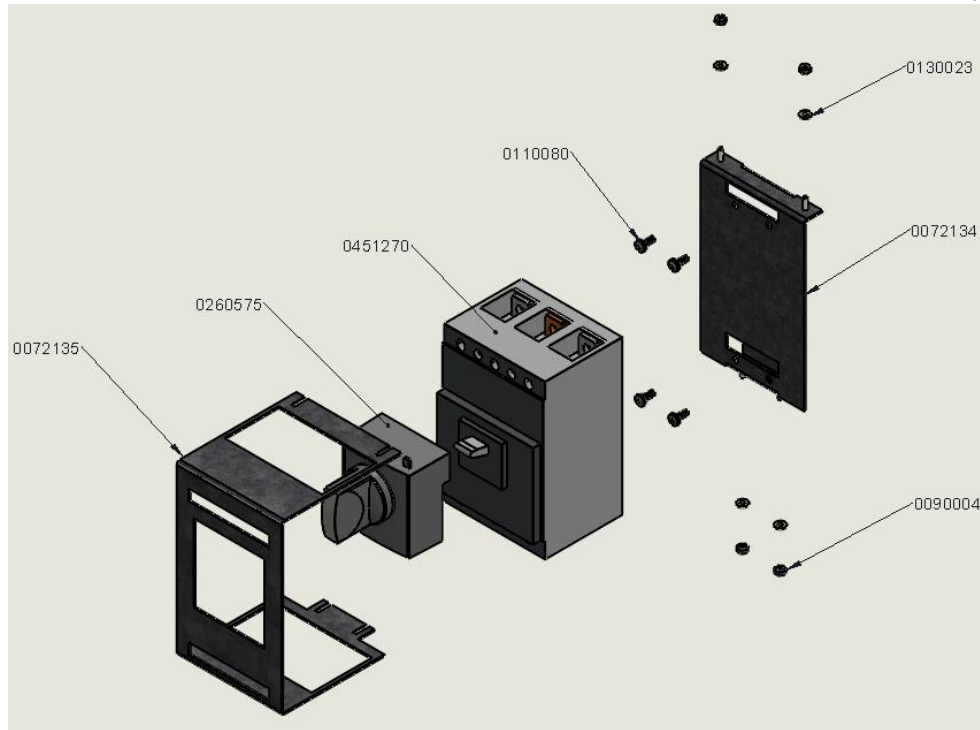
1030370_04

- **Circuit Breaker, ABB # XT4NU3150AFF000XXX, TEAL P/N 4990992-01**
Reference Designator: CB2
Tools: Phillips head screwdriver

1. Open the front door on the panel
2. Disconnect wires 1/65, 2/66, and 3/67 from the top of the circuit breaker, and wires 41 and 42 from U/V



3. Using a Phillips head screwdriver, remove the screws holding the circuit breaker to the plate
4. Remove the old circuit breaker
5. Install the elevation bracket
6. Using the provided hardware, install the new circuit breaker
7. Install the filler cover bracket in front of the circuit breaker
8. Connect the wires removed in step 2



- **Circuit Breaker, ABB # XT1NU3050AAA000XXX, TEAL P/N 4990932**

Reference Designator: CB3

Tools: Phillips head screwdriver

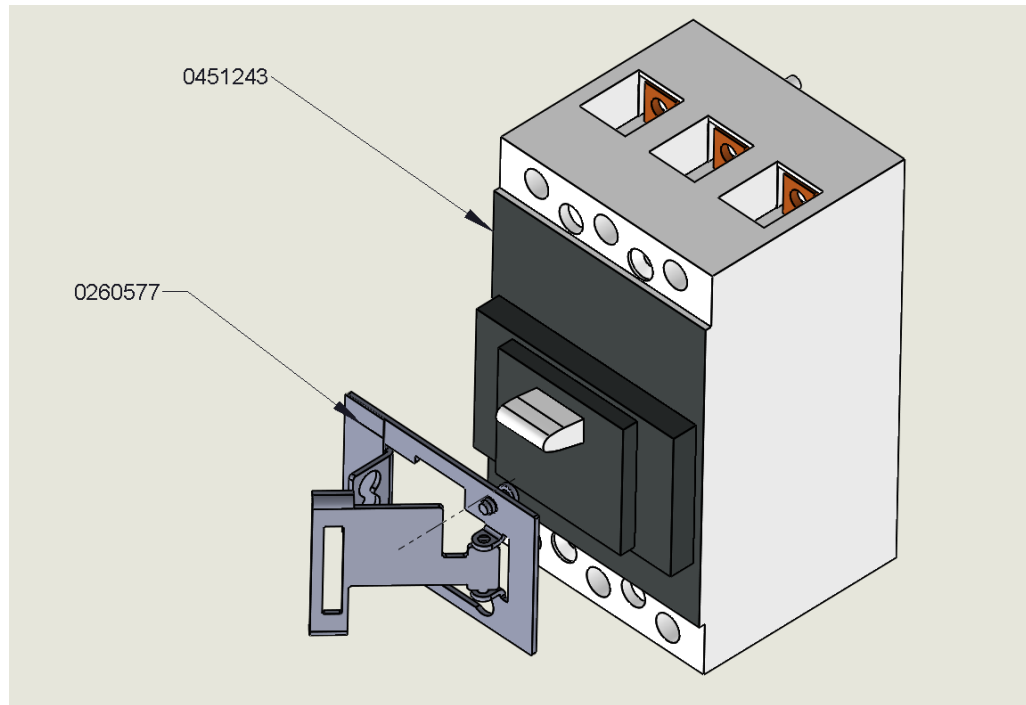
1. Open the front door on the panel
2. Disconnect wires 5, 6, 7, 8, and 9 from the top of the circuit breaker and wires 10, 11, and 12 from the bottom of the circuit breaker





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3. Using a Phillips head screwdriver, remove the screws holding the circuit breaker to the plate
4. Remove the old circuit breaker
5. Install the new circuit breaker
6. Reconnect the wires disconnected in step 2

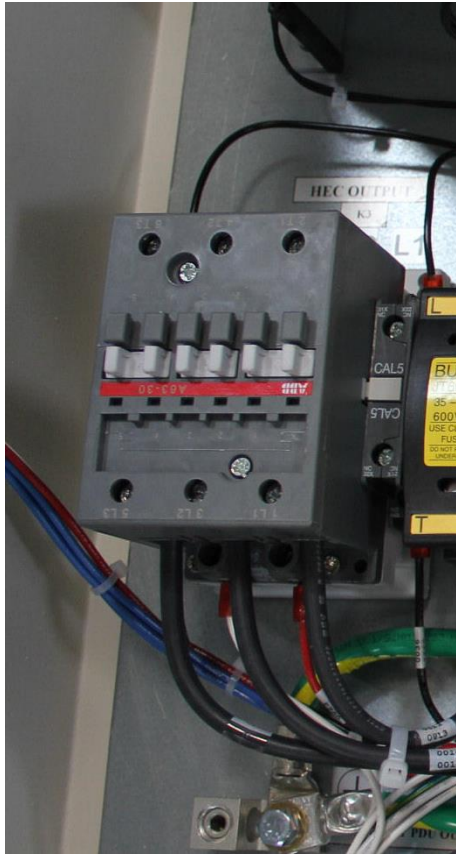


- **Contactor, ABB # AF65-30-11-13, TEAL P/N 4990935**

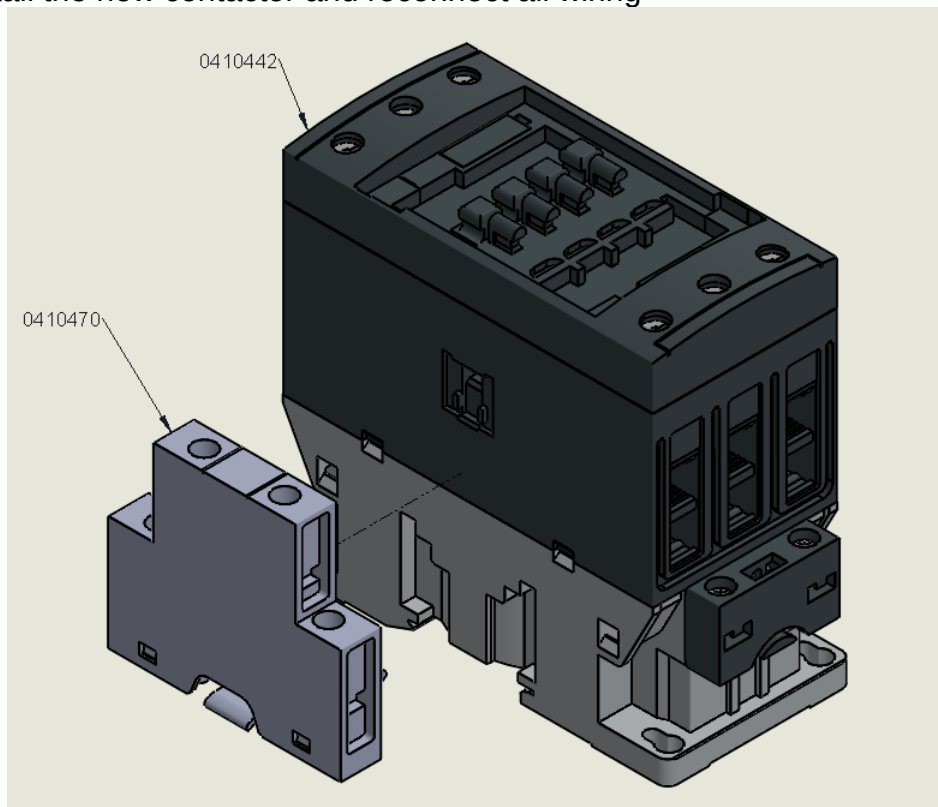
Reference Designator: K3

Tools: Phillips head screwdriver

1. Open the front door on the panel
2. Disconnect wires 13,14, and 15 from the input (L1, L2, L3) of the contactor
3. Disconnect wires 23 and 25 from the coil (A1 and A2) of the contactor
4. Disconnect wires 33 and 34 from AUX1



5. Remove the contactor from the DIN rail
6. Install the new contactor and reconnect all wiring





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6- ENVIRONMENTAL CONDITIONS FOR TRANSPORT AND STORAGE

Ambient Temperature Range:

Non operating, packaged: -40°C to +85°C

Relative Humidity Range:

10 – 90% non-condensing

Atmospheric Pressure Range:

Sea level to 10,000 feet

7- CONTACT INFORMATION

Office Locations:

Main Office:

TEAL Electronics

4350 Executive Drive, Suite 315

San Diego, CA 92121

(800) 888-TEAL or (800) 888-8325

(858) 558-9000

(858) 558-9500 FAX

Office Hours: 8AM to 5PM PST

8- REVISION HISTORY

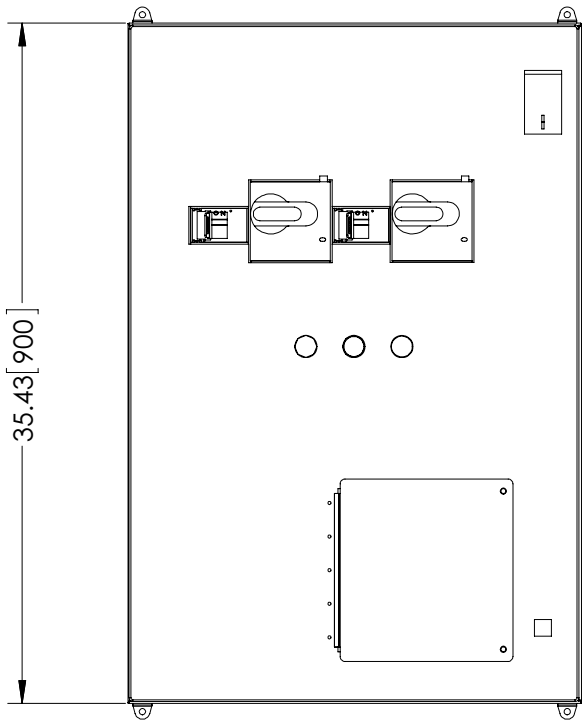
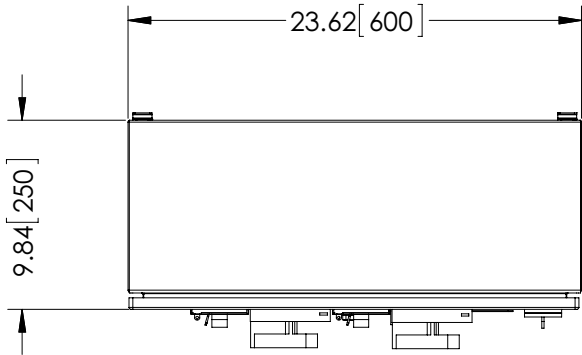
Revision	A0	Initial Release. PDU upgraded to RoHS compliant.	
Approved By:	T. Cole		Date: 2/21/14
Revision	A1	As per ECR 16010, section 5-Maintenance changed to "Annually, the unit should be serviced by a qualified electrician to tighten all electrical connections" per customer request.	
Approved By	T. Cole		Date: 2/8/16
Revision	02	As per ECR M11422, updated FRU kit list and FRU kit instructions, updated address	
Approved By	S. Woods		Date: 1/8/18



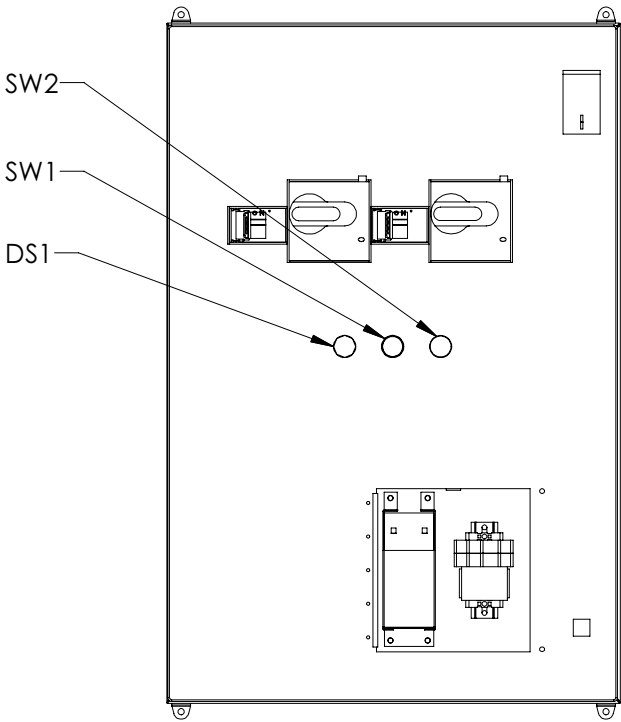
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Revision	03	Updated schematic diagram	
Approved By		S. Woods	Date: 2/11/19
Revision	04	Added dip switch settings image for new breaker	
Approved By		S. Woods	Date: 5/8/19

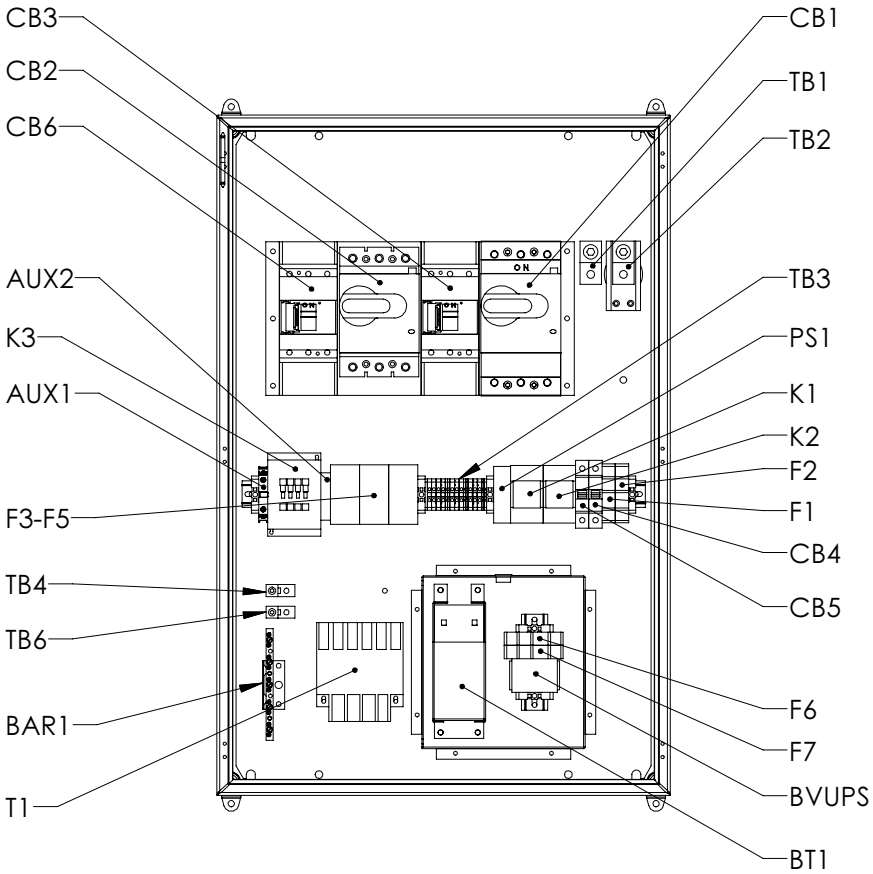
REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
A0	NEW RELEASE. ECO 6715	10-18-12	B. GLICK



DOOR CLOSED



BATTERY COMPARTMENT
PANEL OPEN



DOOR REMOVED

NOTES: (UNLESS OTHERWISE SPECIFIED)



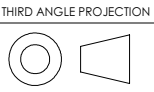
THIS DOCUMENT CONTAINS CONFIDENTIAL PROPRIETARY INFORMATION AND IS NOT TO BE DISTRIBUTED IN WHOLE OR IN PART WITHOUT THE WRITTEN PERMISSION OF TEAL ELECTRONICS CORPORATION.

TOLERANCES, UNLESS OTHERWISE SPECIFIED:
FRACTIONS DECIMALS ANGLES
1/32 .X ±.1 1/2°
XXX ±.015

DIMENSION/COMPONENT DIAGRAM

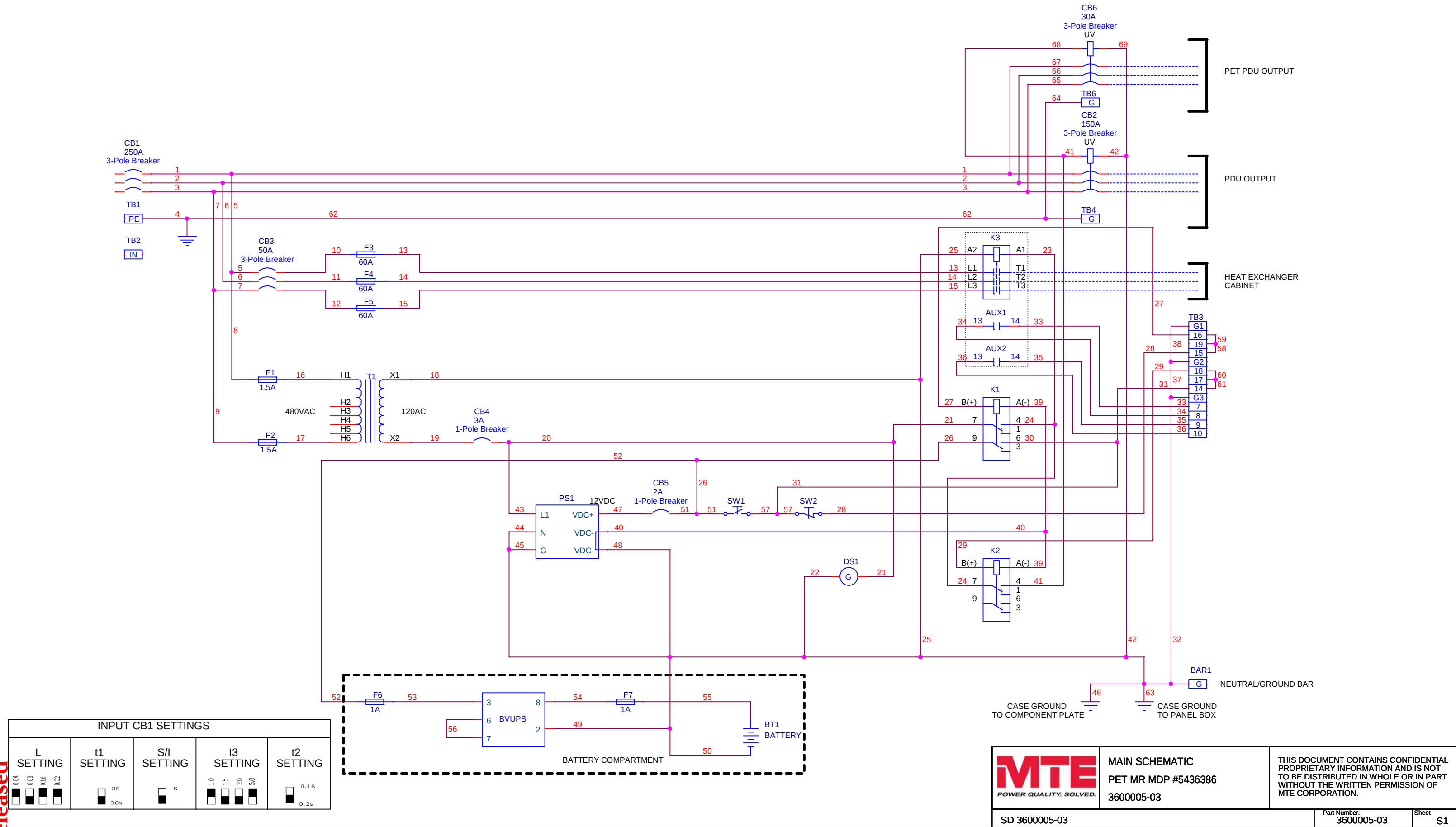
MATERIAL
FINISH

PART NUMBER	9020118	SHEET	1 OF 1	REVISION	A0
FILENAME	9020118_A0				



Released

Revision	Date / By	Description	Approved
00	5-18-2018	INITIAL RELEASE FROM 3600005-02 WITH ABB EOL UPDATES	T. Cole
01	1-22-2019	UPDATED TO REFLECT INPUT BREAKER SWITCH CHANGES	R. Hombsch



PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT****POWER SUPPLY**

- 100-240V Wide Range Input
- NEC Class 2 Compliant
- Adjustable Output Voltage
- Efficiency up to 83.6%
- Compact Design, Width only 22.5mm
- Full Output Power Between -10°C and +60°C
- Large International Approval Package
- 3 Year Warranty

1. GENERAL DESCRIPTION

A compact size, light weight, simple mounting onto the DIN-rail and the utilization of only quality components are what makes the MiniLine power supplies so easy to use and install within seconds.

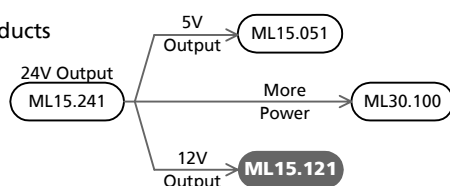
A rugged electrical and mechanical design as well as a high immunity against electrical disturbances on the mains provides reliable output power. This offers superior protection for equipment which is connected to the public mains network or is exposed to a critical industrial environment.

The MiniLine series offers output voltages from 5 to 56Vdc and a power rating from 15W to 100W.

The supplementary MiniLine decoupling diode module MLY02.100 allows building of redundant systems or to protect against back-feed voltages.

ML15

Related products

**2. SHORT-FORM DATA**

Output voltage	DC 12V	
Adjustment range	12-15V	
Output current	1.3-1.0A	12-15V
Output power	15W	
Output ripple	< 75mVpp	20Hz to 20MHz
Input voltage	AC 100-240V	Wide Range Input
Mains frequency	50-60Hz	±6%
AC Input current	typ. 0.28 / 0.17A	At 120 / 230Vac
Power factor	typ. 0.51 / 0.44	At 120 / 230Vac
AC Inrush current	16 / 31A	Typ. peak value at 120 / 230Vac 40°C and cold start
DC Input	85-375Vdc	
Efficiency	typ. 83.0 / 83.6%	At 120 / 230Vac
Losses	typ. 3.2 / 3.1W	At 120 / 230Vac
Temperature range Derating	-10°C to +70°C	Operational +60 to +70°C
Hold-up time	typ. 46 / 191ms	At 120 / 230Vac
Dimensions	22.5x75x91mm	WxHxD

3. ORDER NUMBERS

Power supply	ML15.121	Standard unit
Accessory	MLY02.100	Decoupling / Redundancy module

4. MARKINGS

Dec. 2007 / Rev. 1.1 DS-ML15.121-EN

All parameters are specified at 12V, 1.3A, 230Vac, 25°C ambient and after a 5 minutes run-in time unless otherwise noted.

www.pulspower.com Phone +49 89 9278 0 Germany

PULS

MiniLine

ML15.121

12V, 1.3A, SINGLE PHASE INPUT

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INTENDED USE

The power supply shall only be installed and put into operation by qualified personnel.

This power supply is designed for installation in an enclosure and is intended for the general use, such as in industrial control, office, communication, and instrumentation equipment. Do not use this device in aircraft, trains and nuclear equipment, where malfunctioning of the power supply may cause severe personal injury or threaten human life.

TERMINOLOGY AND ABBREVIATIONS

PE and  symbol	PE is the abbreviation for P rotective E arth and has the same meaning as the symbol  .
Earth, Ground	This document uses the term "earth" which is the same as the U.S. term "ground".
T.b.d.	To be defined, value or description will follow later.
AC 230V	A figure displayed with the AC or DC before the value represents a nominal voltage with standard tolerances (usually $\pm 15\%$) included. E.g.: DC 12V describes a 12V battery regardless whether it is charged (13.7V) or discharged (10V). As long as not otherwise stated, AC 100V and AC 230V parameters are valid at 50Hz and AC 120V parameters are valid at 60Hz mains frequency.
230Vac	A figure with the unit (Vac) at the end is a momentary figure without any additional tolerances included.

DISCLAIMER

The information presented in this document is believed to be accurate and reliable and may change without notice.

The housing is patent by PULS (US patent No US D442,923S)

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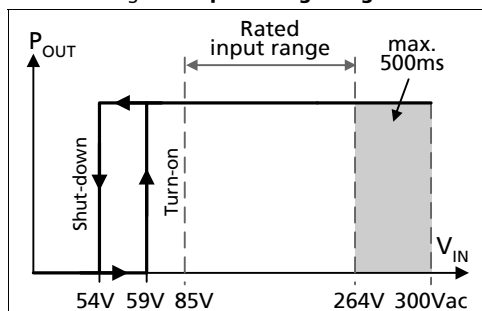
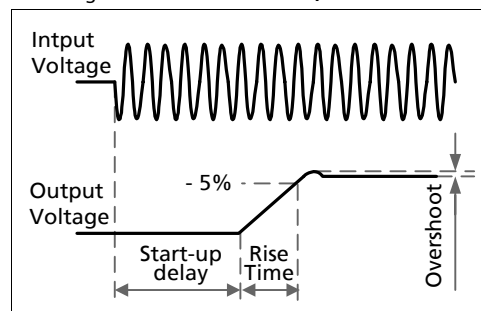
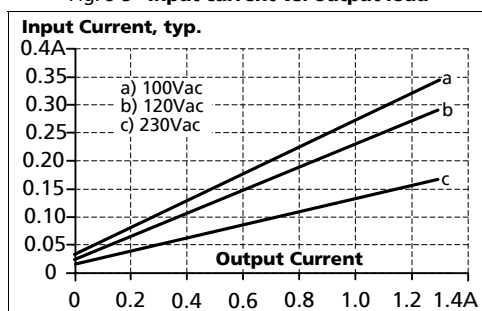
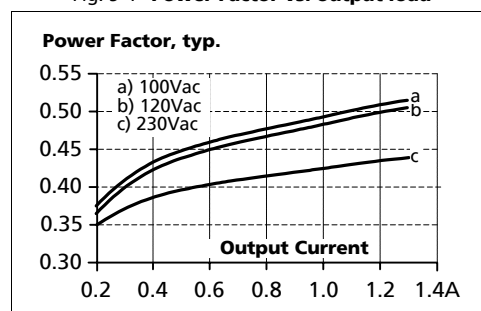
PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT****5. AC-INPUT**

AC input	nom.	AC 100-240V	Wide-range input, TN-, TT-, IT-Mains, see Fig. 5-1
AC input range		85-264Vac 264-300Vac	Continuous operation < 0.5s
Input frequency	nom.	50 – 60Hz	±6%
Turn-on voltage	typ.	59Vac	Steady-state value, see Fig. 5-1
Shut-down voltage	typ.	54Vac	Steady-state value, see Fig. 5-1

		AC 100V	AC 120V	AC 230V	
Input current (rms)	typ.	0.34A	0.28A	0.17A	At 12V, 1.3A see Fig. 5-3
Power factor *)	typ.	0.52	0.51	0.44	At 12V, 1.3A see Fig. 5-1
Crest factor **)	typ.	3.45	3.53	3.94	At 12V, 1.3A
Start-up delay	typ.	780ms	780ms	780ms	See Fig. 5-2
Rise time	typ.	12ms	12ms	12ms	At 12V, 1.3A, see Fig. 5-2
Turn-on overshoot	max.	100mV	100mV	100mV	See Fig. 5-2

*) The power factor is the ratio of the true (or real) power to the apparent power in an AC circuit.

**) The crest factor is the mathematical ratio of the peak value to RMS value of the input current waveform.

Fig. 5-1 Input voltage range**Fig. 5-2 Turn-on behavior, definitions****Fig. 5-3 Input current vs. output load****Fig. 5-4 Power Factor vs. output load**

PULS

MiniLine

ML15.121

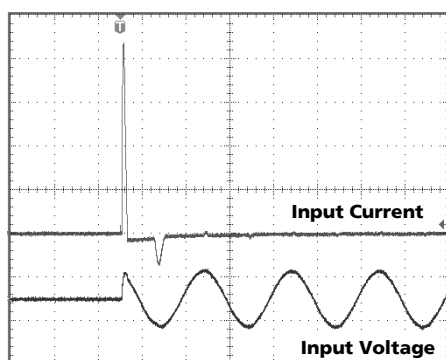
12V, 1.3A, SINGLE PHASE INPUT

6. INPUT INRUSH CURRENT SURGE

A NTC limits the input inrush current after turn-on of the input voltage. The inrush current is input voltage and ambient temperature dependent.

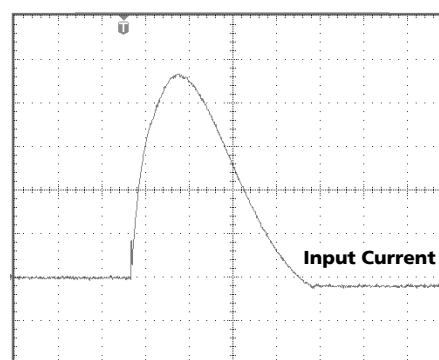
		AC 100V	AC 120V	AC 230V	
Inrush current	max.	13A _{peak}	16A _{peak}	31A _{peak}	40°C ambient, cold start
	typ.	11A _{peak}	13A _{peak}	26A _{peak}	40°C ambient, cold start
Inrush energy	typ.	0.1A ² s	0.1A ² s	0.4A ² s	40°C ambient, cold start

Fig. 6-1 Input inrush current, typical behavior



Input: 230Vac
Output: 12V, 1.3A
Ambient: 25°C
Upper curve: Input current 5A / DIV
Lower curve: Input voltage 500V / DIV
Time scale: 10ms / DIV

Fig. 6-2 Input inrush current, zoom into the first peak



Input: 230Vac
Output: 12V, 1.3A
Ambient: 25°C
Input current curve: 5A / DIV, 500µs / DIV
I_{peak} 23A

The charging current into EMI suppression capacitors is disregarded in the first microseconds after switch-on.

7. HOLD-UP TIME

		AC 100V	AC 120V	AC 230V	
Hold-up Time	typ.	62ms	95.5ms	365ms	12V, 0.65A, see Fig. 7-1
	typ.	30ms	46ms	191ms	12V, 1.3A see Fig. 7-1

Fig. 7-1 Hold-up time vs. input voltage

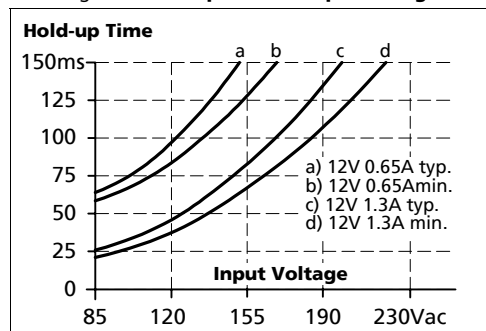
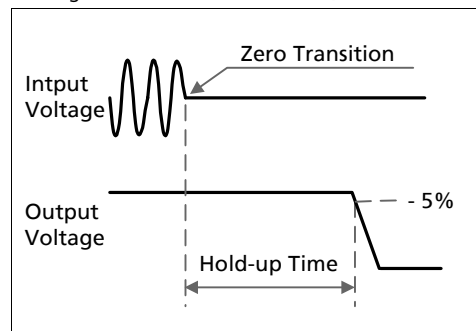


Fig. 7-2 Shut-down behavior, definitions



Note: At no load, the hold-up time can be up to several seconds. The green DC-ok lamp is also on during this time.

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All parameters are specified at 12V, 1.3A, 230Vac, 25°C ambient and after a 5 minutes run-in time unless otherwise noted.

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PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT**

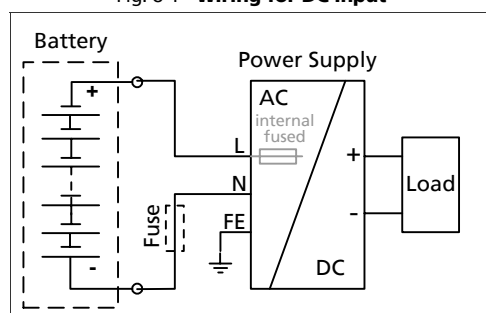
8. DC-INPUT

DC input	nom.	DC 110-290V	-25%/+30%
DC input range	min.	85-375Vdc	Continuous operation
DC input current	typ.	0.16A / 0.057A	110Vdc / 300Vdc, at a load of 12V and 1.3A
Turn-on voltage	typ.	80Vdc	Steady state value
Shut-down voltage	typ.	60Vdc	Steady state value

Instructions for DC use:

- Use a battery or similar DC source.
- Connect +pole to L and – pole to N.
- In case the – pole of the battery is not connected to earth, use an appropriate fuse to protect the N terminal.

Fig. 8-1 **Wiring for DC Input**



PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT****9. OUTPUT**

Output voltage	nom.	12V	
Adjustment range	min.	12-15V	Guaranteed
	max.	16V	At clockwise end position of potentiometer
Factory setting		12.0V	±0.2%, at full load, cold unit
Line regulation	max.	10mV	85 to 264Vac
Load regulation	max.	100mV	Static value, 0A → 1.3A → 0A
Ripple and noise voltage	max.	75mVpp	20Hz to 20MHz, 50Ohm
Output capacitance	typ.	1650μF	
Output current	nom.	1.3A	At 12V, see Fig. 9-1 for typical figures
	nom.	1.0A	At 15V, see Fig. 9-1 for typical figures
Output power	nom.	15W	
Short-circuit current	min.	Hiccup mode	See Fig. 9-2
	max.	Hiccup mode	See Fig. 9-2

Fig. 9-1 Output voltage vs. output current, typ.

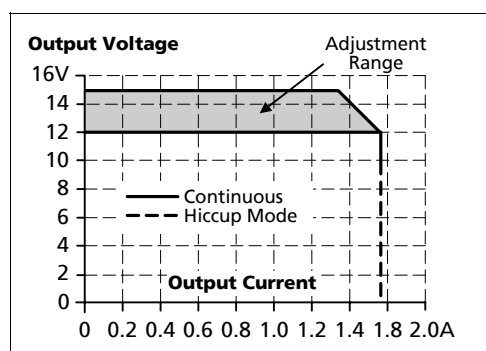
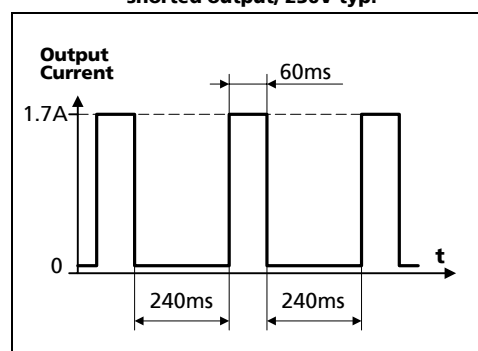


Fig. 9-2 Hiccup mode, Output current at shorted output, 230V typ.

**Peak current capability (up to several ms)**

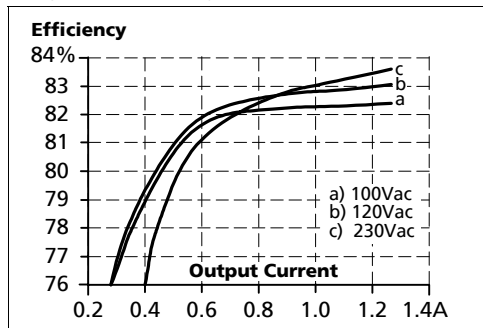
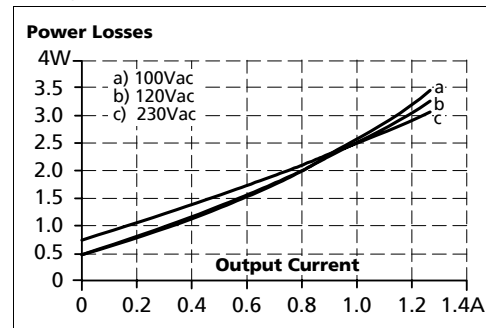
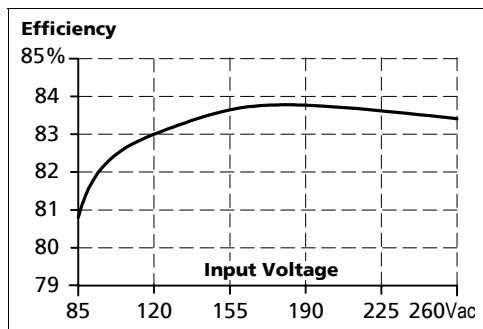
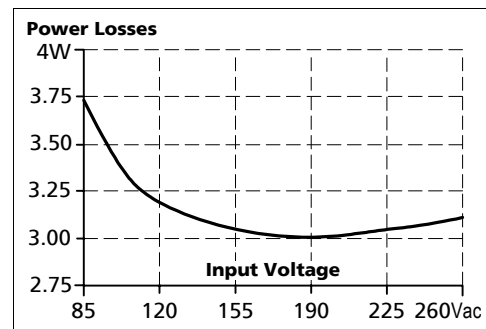
The power supply can deliver a peak current which is higher than the specified short term current. This helps to start current demanding loads or to safely operate subsequent circuit breakers.

The extra current is supplied by the output capacitors inside the power supply. During this event, the capacitors will be discharged and cause a voltage dip on the output. Detailed curves can be found in chapter 25.1

Peak current voltage dips	typ.	from 12V to 7.8V	At 2.6A for 50ms, resistive load
	typ.	from 12V to 5.2V	At 6.5A for 2ms, resistive load
	typ.	from 12V to 2.0V	At 6.5A for 5ms, resistive load

PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT****10. EFFICIENCY AND POWER LOSSES**

		AC 100V	AC 120V	AC 230V	
Efficiency	typ.	82.4%	83%	83.6%	1.3A, 12V (full load)
Power losses	typ.	0.5W	0.5W	0.8W	At no load
	typ.	1.55W	1.55W	1.75W	0.65A, 12V (half load)
	typ.	3.4W	3.2W	3.1W	1.3A, 12V (full load)

Fig. 10-1 Efficiency vs. output current at 12V**Fig. 10-2 Losses vs. output current at 12V****Fig. 10-3 Efficiency vs. input voltage at 12V and 1.3A****Fig. 10-4 Losses vs. input voltage at 12V and 1.3A**

PULS

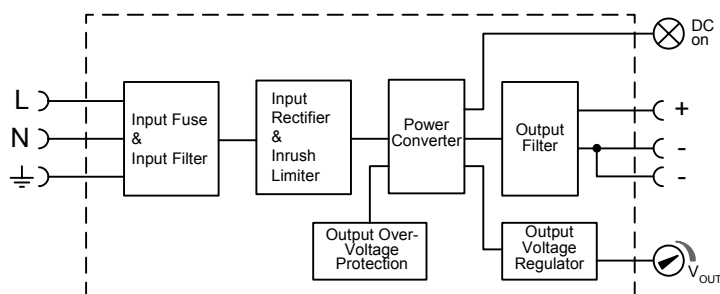
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12V, 1.3A, SINGLE PHASE INPUT

11. FUNCTIONAL DIAGRAM

Fig. 11-1 Functional diagram



12. RELIABILITY

These units are extremely reliable and use only the highest quality materials. The number of critical components such as electrolytic capacitors has been reduced.

		AC 100V	AC 120V	AC 230V	
Lifetime expectancy	min.	123 000h	126 000h	125 000h	40°C, 12V, 1.3A
	min.	> 15 years	> 15 years	> 15 years	40°C, 12V, 0.65A
	min.	> 15 years	> 15 years	> 15 years	25°C, 12V, 1.3A
MTBF SN 29500, IEC 61709		3 435 000h	3 723 000h	3 811 000h	40°C, 12V, 1.3A
		5 633 000h	6 106 000h	6 205 000h	25°C, 12V, 1.3A
MTBF MIL HDBK 217F		1 482 000h	1 534 000h	1 451 000h	40°C, 12V, 1.3A, Ground Benign GB40
		1 986 000h	2 056 000h	1 944 000h	25°C, 12V, 1.3A, Ground Benign GB25

The **Lifetime expectancy** shown in the table indicates the operating hours (service life) and is determined by the lifetime expectancy of the built-in electrolytic capacitors.

Lifetime expectancy is specified in operational hours. Lifetime expectancy is calculated according to the capacitor's manufacturer specification. The prediction model allows a calculation of up to 15 years from date of shipment.

MTBF stands for **Mean Time Between Failure**, which is calculated according to statistical device failures, and indicates reliability of a device. It is the statistical representation of the likelihood of a unit to fail and does not necessarily represent the life of a product.

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12V, 1.3A, SINGLE PHASE INPUT

13. FRONT SIDE AND USER ELEMENTS

Fig. 13-1 Front side



A Output Terminals

Screw terminals

Dual terminals for the negative pole allows an easy earthing of the output voltage

+ Positive output

- Negative (return) output

B Input Terminals

Screw terminals

L Neutral input

N Line (hot) input



EMI ground

Ground this terminal to minimize high-frequency emissions.

For safety reasons, connecting to ground is not required.

C DC-ON LED (green)

On when the voltage is > 10.5V

D Output voltage potentiometer

Turn to set the output voltage. Factory setting is 12.0V.

14. TERMINALS AND WIRING

All terminals are easy to access when mounted on the panel. Input and output terminals are separated from each other (input below, output above) to help in error-free wiring.

Type	Screw terminals
Solid wire	0.5-6mm ²
Stranded wire	0.5-4mm ²
American wire gauge	20-10 AWG
Ferrules	Allowed, but not required
Recommended tightening torque	1Nm, 9lb.in
Screwdriver	3.5mm slotted or Pozidrive No 2
Wire stripping length	7mm / 0.275inch

Instructions:

- Use appropriate copper cables that are designed for an operating temperatures of 60°C (for ambient up to 45°C) and 75°C (for ambient up to 60°C), minimum.
- Ferrules are allowed, but not required.
- Follow local and national installation codes and regulations!
- Ensure that all strands of a stranded wire enter the terminal connection!
- In order to fulfill GL requirements, unused terminal spaces must be closed.

PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT****15. EMC**

The CE mark indicates conformance with EMC directive 89/336/EC, 93/68/EC and 2004/108/EC and the low-voltage directive (LVD) 73/23/EC, 93/68/EC, 2006/95/EC. A detailed EMC Report is available on request.

All results and investigation were performed with functional earth wire.

EMC Immunity	EN 61000-6-2, EN 61000-6-1	Generic standards		
Electrostatic discharge	EN 61000-4-2	Contact discharge	8kV	Criterion A
		Air discharge	8kV	Criterion A
Electromagnetic RF field	EN 61000-4-3	80MHz-2.7GHz	10V/m	Criterion A
Fast transients (Burst)	EN 61000-4-4	Input lines	4kV	Criterion A
		Output lines	2kV	Criterion A
Surge voltage on input	EN 61000-4-5	L → N	2kV	Criterion A
		N / L → PE	4kV	Criterion A
Surge voltage on output	EN 61000-4-5	+ → -	500V	Criterion A
		+ / - → PE	500V	Criterion A
Conducted disturbance	EN 61000-4-6	0,15-80MHz	10V	Criterion A
Mains voltage dips	EN 61000-4-11	0% of 100Vac	0Vac, 20ms	Criterion A
		40% of 100Vac	40Vac, 200ms	Criterion C
		70% of 100Vac	70Vac, 500ms	Criterion A
		0% of 200Vac	0Vac, 20ms	Criterion A
		40% of 200Vac	80Vac, 200ms	Criterion A
		70% of 200Vac	140Vac, 500ms	Criterion A
Voltage interruptions	EN 61000-4-11		0Vac, 5000ms	Criterion C
Input voltage swells	PULS internal standard		300Vac, 500ms	Criterion A
Powerful transients	VDE 0160	Over entire load range	750V, 1.3ms	Criterion A

Criteria:

A: Power supply shows normal operation behavior within the defined limits.

C: Temporary loss of function is possible. Power supply might shut-down and restarts by itself. No damages or hazards to the power supply will occur.

EMC Emission	EN 61000-6-3, EN 61000-6-4	Generic standards	
Conducted emission	EN 55011, EN 55022, FCC Part 15, CISPR 11, CISPR 22 EN 55022	Class B, input lines	
		Class B, output lines	
Radiated emission	EN 55011, EN 55022	Class B	
Harmonic input current	EN 61000-3-2	Not applicable (below 75W input power)	
Voltage fluctuations, flicker	EN 61000-3-3	Fulfilled	

This device complies with FCC Part 15 rules.

Operation is subjected to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

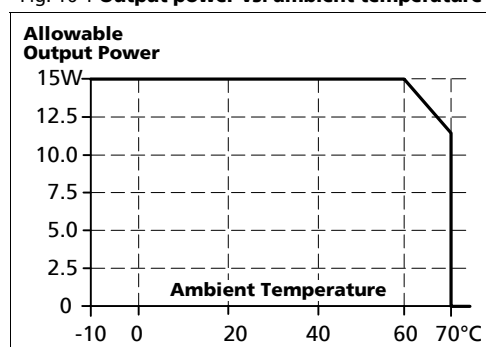
Switching frequency constant, typ. 65kHz

PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT**

16. ENVIRONMENT

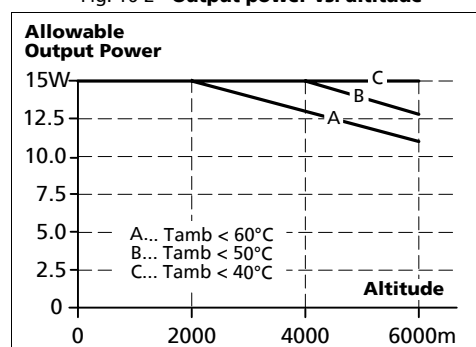
Operational temperature	-10°C to +70°C (14°F to 158°F)	Reduce output power acc. to Fig. 16-1
Output de-rating	0.4W/°C	60-70°C (140°F to 158°F), see Fig. 16-1
Storage temperature	-40 to +85°C (-40°F to 185°F)	Storage and transportation
Humidity	5 to 95% r.H.	IEC 60068-2-30 Do not energize while condensation is present
Vibration sinusoidal	2-17.8Hz: ±1.6mm; 17.8-500Hz: 2g 2 hours / axis	IEC 60068-2-6
Shock	15g 6ms, 10g 11ms 3 bumps / direction, 18 bumps in total	IEC 60068-2-27
Altitude	0 to 6000m (0 to 20000ft)	Reduce output power or ambient temperature above 2000m sea level. Above 2000m (6500ft), see Fig. 16-2
Output de-rating (for altitude)	1W/1000m or 5°C/1000m	
Over-voltage category	III II	EN 50178, IEC 62103, up to 2000m Altitudes from 2000m to 6000m
Degree of pollution	2	EN 50178, IEC 62103, non conductive

Fig. 16-1 Output power vs. ambient temperature



The ambient temperature is defined 2cm below the unit.

Fig. 16-2 Output power vs. altitude



17. PROTECTION FEATURES

Output protection	Electronically protected against overload, no-load and short-circuits	
Output over-voltage protection	Typ. 21.5V, max. 25V	In case of an internal power supply fault, a redundant circuit limits the maximum output voltage. In such a case, the output shuts down and stays down until the input voltage is turned off and on again
Output over-current protection	Electronically limited	See Fig. 9-1
Degree of protection	IP 20	EN/IEC 60529
Penetration protection	> 2.5mm in diameter	E.g. screws, small parts
Over-temperature protection	Not included	
Input transient protection	MOV	Metal Oxide Varistor
Internal input fuse	T3.15A H.B.C.	Not user replaceable

Note: In case of a over-voltage, overload, no-load or short-circuit event, an audible noise may be heard.

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18. SAFETY

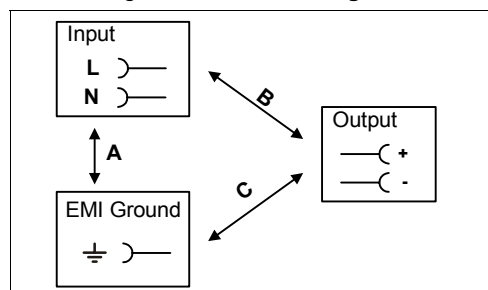
Input / output separation	SELV PELV double or reinforced insulation	IEC/EN 60950-1 EN 60204-1, EN 50178, IEC 60364-4-41, IEC 62103
Class of protection	II	
Isolation resistance	> 5MΩ	Input to output, 500Vdc
Touch current (leakage current)	typ. 0.16mA ¹⁾ / 6μA ²⁾ typ. 0.25mA ¹⁾ / 7μA ²⁾ typ. 0.38mA ¹⁾ / 10μA ²⁾ < 0.20mA ¹⁾ / 7μA ²⁾ < 0.30mA ¹⁾ / 8μA ²⁾ < 0.53mA ¹⁾ / 12μA ²⁾	100Vac, 50Hz, TN mains 120Vac, 60Hz, TN mains 230Vac, 50Hz, TN mains 110Vac, 50Hz, TN mains 132Vac, 60Hz, TN mains 264Vac, 50Hz, TN mains

1) Functional earth terminal connected to Protective earth

2) Functional earth terminal not connected

19. DIELECTRIC STRENGTH

Fig. 19-1 Dielectric strength



		A	B	C
Type test	60s	2500Vac	3000Vac	500Vac
Factory test	5s	2500Vac	2500Vac	500Vac
Field test	5s	2000Vac	2000Vac	500Vac

Type tests and factory tests:

Conducted by the manufacturer. Do not repeat test in field!

Rules for field test:

Use appropriate test equipment which applies the voltage with a slow ramp! Connect L and N together as well as all output poles.

The output voltage is floating and has no ohmic connection to ground.

To fulfill the PELV requirements according to EN60204-1 § 6.4.1, it is recommended that either the + pole, the – pole or any other part of the output circuit should be connected to the protective earth system. This helps to avoid situations in which a load starts unexpectedly or can not be switched off when an unnoticed earth faults occur.

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20. APPROVALS

IEC 60950-1	IECEE CB SCHEME	CB Scheme, Information Technology Equipment
UL 508		LISTED E198865 Industrial Control Equipment
UL 60950-1		RECOGNIZED E137006 recognized for the use in U.S.A. (UL 60950-1) and Canada (C22.2 No. 60950). Information Technology Equipment, Level 3
NEC Class 2		According to NEC (National Electrical Code) Article 725-41 (4). Listed as Limited Power Source (LPS) in the UL 60950-1 UL report.
Class I Div 2		RECOGNIZED E246877 Hazardous Location Class I Div 2 T4 Groups A,B,C and D ANSI/ISA-12.12.01 (UL 1604) The unit is suitable for use in Class I Division 2 Groups A, B, C, D locations. Substitution of components may impair suitability for Class I Division 2 environment. Do not disconnect equipment unless power has been switched off. Wiring must be in accordance with Class I, Division 2 wiring methods of the National Electrical Code, NFPA 70, and in accordance with other local or national codes.
CSA		CSA approval for Canada CAN/CSA C22.2 No 107-1; CAN/ CSA 60950-1-03; UL60950-1
Marine		GL (Germanischer Lloyd) classified and ABS (American Bureau for Shipping) PDA for marine and offshore applications. Environmental category: C, EMC2

21. FULFILLED STANDARDS

EN 61558-2-17	Safety of Power Transformers
EN/IEC 60204-1	Safety of Electrical Equipment of Machines
EN/IEC 61131-2	Programmable Controllers
EN 50178, IEC 62103	Electronic Equipment in Power Installations

22. USED SUBSTANCES

Electrolytic capacitors included in this unit do not use electrolytes such as Quaternary Ammonium Salt Systems.

The unit conforms to the RoHS directive 2002/96/EC

Plastic housings and other molded plastic materials are free of halogens, wires and cables are not PVC insulated.

The materials used in our production process do not include the following toxic chemicals:

Polychlorinated Biphenyl (PCB), Pentachlorophenol (PCP), Polychlorinated naphthalene (PCN), Polybrominated Biphenyl (PBB), Polybrominated Biphenyl Oxide (PBO), Polybrominated Diphenyl Ether (PBDE), Polychlorinated Diphenyl Ether (PCDE), Polybrominated Diphenyl Oxide (PBDO), Cadmium, Asbestos, Mercury, Silica

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23. PHYSICAL DIMENSIONS AND WEIGHT

Weight	175g / 0.39lb
DIN-Rail	Use 35mm DIN-rails according to EN 60715 or EN 50022 with a height of 7.5 or 15mm. The DIN-rail height must be added to the depth (91mm) to calculate the total required installation depth.
Electronic files with mechanical data can be downloaded at www.pulspower.com	

Fig. 23-1 Front view

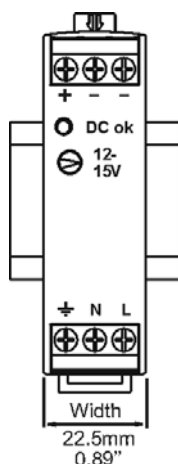
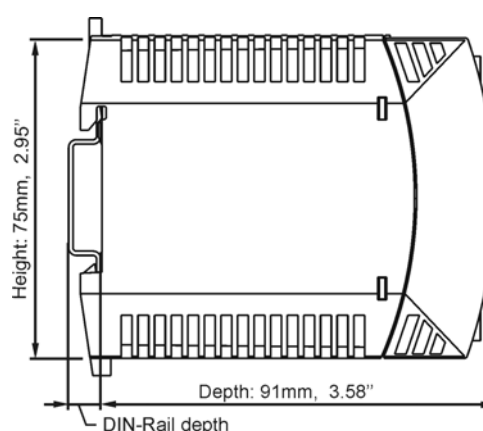


Fig. 23-2 Side view



24. INSTALLATION AND OPERATION INSTRUCTIONS

Mounting:

Output terminal must be located on top and input terminal on the bottom. For other orientations see chapter 25.12. An appropriate electrical and fire end-product enclosure should be considered in the end use application.

Cooling:

Convection cooled, no forced cooling required. Do not cover ventilation grill (e.g. cable conduits) by more than 30%!

Installation clearances:

25mm on top and bottom.

Risk of electrical shock, fire, personal injury or death!

Turn power off before working on the power supply. Protect against inadvertent re-powering.

Make sure the wiring is correct by following all local and national codes.

Do not open, modify or repair the unit.

Use caution to prevent any foreign objects from entering into the housing.

Do not use in wet locations or in areas where moisture or condensation can be expected.

Service parts:

The unit does not contain any service parts. The tripping of an internal fuse is caused by an internal fault. If damage or malfunctioning should occur during operation, immediately turn power off and send unit to the factory for inspection!

PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT**

25. APPLICATION NOTES

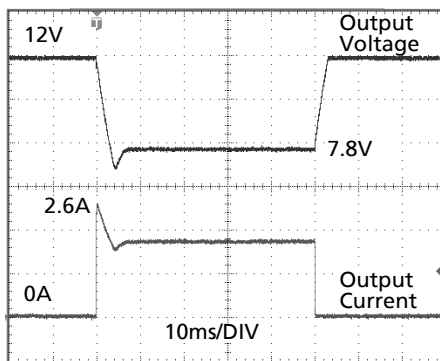
25.1. PEAK CURRENT CAPABILITY

Solenoids, contactors and pneumatic modules often have a steady state (sealed) coil and a pick-up coil. The inrush current demand of the pick-up coil is several times higher than the steady state current and usually exceeds the nominal output current. The same situation applies, when starting a capacitive load.

Branch circuits are often protected with circuit breakers or fuses. In case of a short or an overload in the branch circuit, the protective device (fuse, circuit breaker) needs a certain amount of over-current to trip or to blow. The peak current capability ensures the safe operation of subsequent circuit breakers.

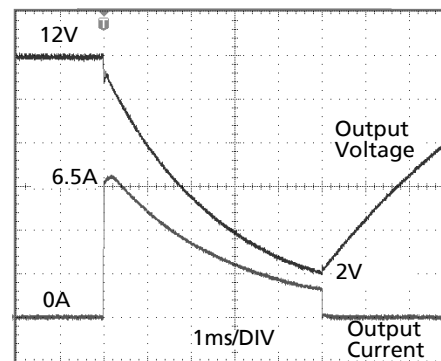
Assuming the input voltage is turned on before such an event, the built-in large sized output capacitors inside the power supply can deliver extra current. Discharging this capacitor causes a voltage dip on the output. The following two examples show typical voltage dips:

Fig. 25-1 **Peak load 2.6A for 50ms, typ.**



Peak load 2.6A (resistive) for 50ms
Output voltage dips from 12V to 7.8V.

Fig. 25-2 **Peak load 6.5A for 5ms, typ.**



Peak load 3.15A (resistive) for 5ms
Output voltage dips from 12V to 2V.

25.2. BACK-FEEDING LOADS

Loads such as decelerating motors and inductors can feed voltage back to the power supply. This feature is also called return voltage immunity or resistance against Back- E.M.F. (Electro Magnetic Force).

The maximum allowed feed back voltage is 25Vdc. The absorbing energy can be calculated according to the built-in large sized output capacitor which is specified in chapter 9.

This power supply is resistant and does not show malfunctioning when a load feeds back voltage to the power supply. It does not matter, whether the power supply is on or off. However, please note that the output voltage can dip to zero for approximately 200ms if the back-feed voltage is removed.

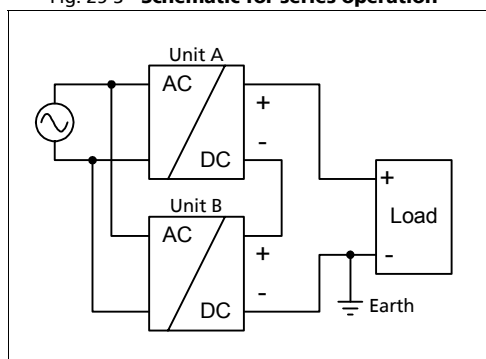
25.3. INDUCTIVE AND CAPACITIVE LOADS

The unit is designed to supply unlimited inductive loads. The max. capacitive load depend on the steady state output current. At 1.3A output current, the output capacity should not be larger than 1000µF at 0.65A output not larger than 2500µF. In case of larger capacitors, the unit can show start-up attempts or start-up problems.

25.4. SERIES OPERATION

The power supply can be put in series to increase the output voltage.

Fig. 25-3 Schematic for series operation



Instructions for use in series:

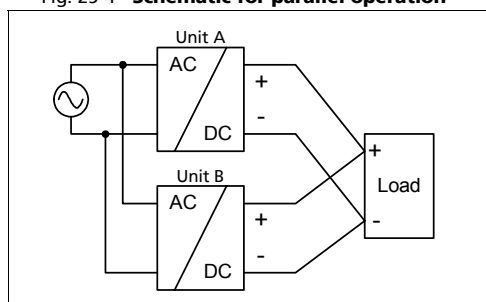
- It is possible to connect as many units in series as needed, providing the sum of the output voltage does not exceed 150Vdc.
- Voltages with a potential above 60Vdc are not SELV any more and can be dangerous. Such voltages must be installed with a protection against touching.
- For serial operation use power supplies of the same type.
- Earthing of the output is required when the sum of the output voltage is above 60Vdc.

Note: Avoid return voltage (e.g. from a decelerating motor or battery) which is applied to the output terminals.

25.5. PARALLEL USE TO INCREASE OUTPUT POWER

Several power supplies can be paralleled to increase the output power. The ML15.121 has no feature included which balances the load current between the power supplies. Usually the power supply with the higher adjusted output voltage draws current until it goes into current limitation. This means no harm to this power supply as long as the ambient temperature stays below 50°C.

Fig. 25-4 Schematic for parallel operation



Instructions for parallel use:

- Use only power supplies from the same series (ML-Series).
- Adjust the output voltages of all power supplies to approximately the same value ($\pm 100\text{mV}$).
- A fuse (or diode) on the output is only required if more than three units are connected in parallel.
- Ensure that the ambient temperature of the power supply does not exceed 50°C.

25.6. PARALLEL USE FOR REDUNDANCY

Power supplies can be paralleled for redundancy to gain a higher system reliability. Redundant systems require a certain amount of extra power to support the load in case one power supply unit fails. The simplest way is to put two MiniLine power supplies in parallel. This is called a 1+1 redundancy. In case one power supply unit fails, the other one is automatically able to support the load current without any interruption. Redundant systems for a higher power demand are usually built in a N+1 method. E.g. Five power supplies, each rated for 1.3A are paralleled to build a 5A redundant system. If one unit fails, the 5A can still be drawn.

Please note: This simple way to build a redundant system has two major disadvantages:

- The faulty power supply can not be recognized. The green LED will still be on since it is reverse-powered from the other power supply.
- It does not cover failures such as an internal short circuit in the secondary side of the power supply. In such a situation the defective unit becomes a load for the other power supplies and the output voltage can not be maintained any more.

This above conditions can be avoided by utilizing decoupling diodes which are included in the decoupling module MLY02.100.

Other recommendations for building redundant power systems:

- a) Use separate input fuses for each power supply.
- b) Set the output voltage of the power supplies approx. 1V higher to compensate for the voltage drop.
- c) When possible, connect each power supply to different phases of the mains network.

25.7. CHARGING OF BATTERIES

The power supply can be used for float-charging of lead-acid or maintenance free 12V VRLA batteries.

Instructions for charging batteries:

- a) Ensure that the ambient temperature of the power supply does not exceed 50°C.
- b) Set the output voltage on a disconnected load, very precisely to the end-of-charge voltage according to the expected battery temperature.

End-of-charge voltage	13.9V	13.75V	13.6V	13.4V
Battery temperature	10°C	20°C	30°C	40°C

- c) Use a 2A or 3A circuit breaker (or blocking diode) between the power supply and the battery.
- d) Ensure that the output current of the power supply is below the allowed charging current of the battery.
- e) Use only matched batteries when putting 6V types in series.
- f) The return current to the power supply is typ. 12mA at 12.5Vdc when the power supply is switched off (except in case a blocking diode is utilized).

25.8. EXTERNAL INPUT PROTECTION

The unit is tested and approved for branch circuits up to 15A (UL) or 16A (IEC). External protection is only required if the supplying branch has an ampacity greater than this. In some countries local regulations might apply so check local codes and requirements.

If an external protective device is utilized, a minimum value is required to avoid undesired tripping of the fuse.

		B-Characteristic	C-Characteristic
Ampacity	max.	15A (UL), 16A (IEC)	15A (UL), 16A (IEC)
	min.	10A	6A

25.9. FUNCTIONAL EARTH TERMINAL

From a safety standpoint, the unit is designed according to the requirements for Protection Class 2 which does not require an earth connection. However, connecting the Functional Earth terminal can be beneficial to gain a high EMI immunity.

Symmetrical spikes or fast transients on the input side can be conducted directly to earth by the built-in filter capacitors. The magnitude of such spikes or fast transients on the output side caused by the input are much smaller compared to not connecting the FE terminal to ground.

Therefore, we recommend to connect the FE terminal too ground.

Fig. 25-5 **Functional earth terminal connected to earth**

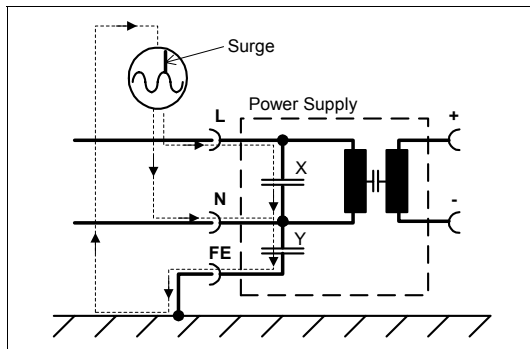
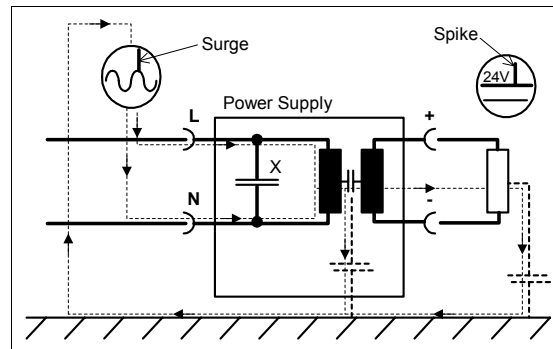


Fig. 25-6 **Functional earth terminal not connected to earth**



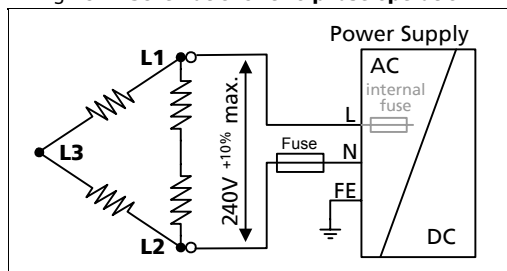
25.10. OPERATION ON TWO PHASES

The power supply can be used on two phases of a three-phase-system

Instructions for two phase operation:

- A phase to phase connection is allowed as long as the supplying voltage is below $240V + 10\%$.
- Use a fuse or a circuit breaker to protect the N input. The N input is internally not protected and is in this case connected to a hot wire. Appropriate fuses or circuit breakers are specified in section 25.7 "External Input Protection".

Fig. 25-7 **Schematic for two phase operation**



PULS**MiniLine****ML15.121****12V, 1.3A, SINGLE PHASE INPUT**

25.11. USE IN A TIGHTLY SEALED ENCLOSURE

When the power supply is installed in a tightly sealed enclosure, the temperature inside the enclosure will be higher than the outside. The inside temperature defines the ambient temperature for the power supply.

Results from such an installation:

Power supply is placed in the middle of the box, no other heat producing equipment inside the box

Enclosure: Rittal Type IP66 Box PK 9510 100, plastic, 130x130x75mm

Input: 230Vac

Load: 12V, 1.3A; load is placed outside the box

Temperature inside the box: 38.9°C (in the middle of the right side of the power supply with a distance of 1cm)

Temperature outside the box: 28.0°C

Temperature rise: 10.9K

Load: 12V, 1.05A; (=80%) load is placed outside the box

Temperature inside the box: 35.9°C (in the middle of the right side of the power supply with a distance of 1cm)

Temperature outside the box: 27.4°C

Temperature rise: 8.5K

25.12. MOUNTING ORIENTATIONS

Mounting orientations other than input terminals on the bottom and output on the top requires a reduction in continuous output power or a limitation in the maximum allowed ambient temperature. The amount of reduction influences the lifetime expectancy of the power supply. Therefore, two different derating curves for continuous operation can be found below:

Curve A1 Recommended output power.

Curve A2 Max allowed output power (results approx. in half the lifetime expectancy of A1).

Fig. 25-8

Mounting Orientation A
Standard Orientation

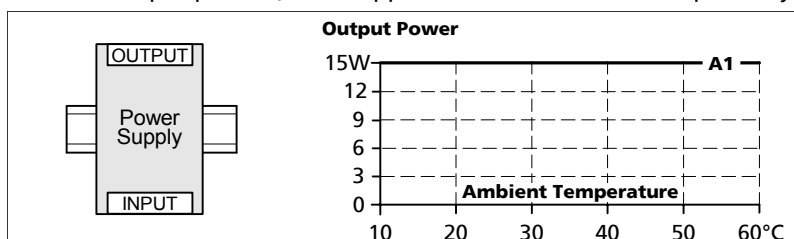


Fig. 25-9

Mounting Orientation B
(Upside down)

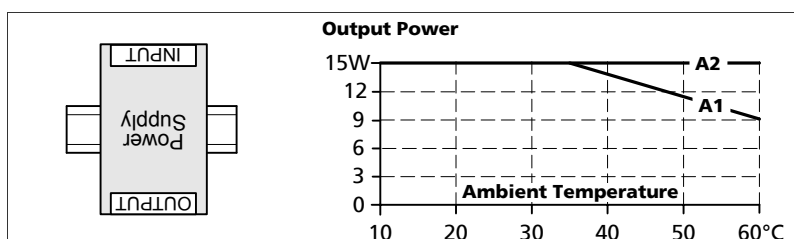


Fig. 25-10

Mounting Orientation C
(Table-top mounting)

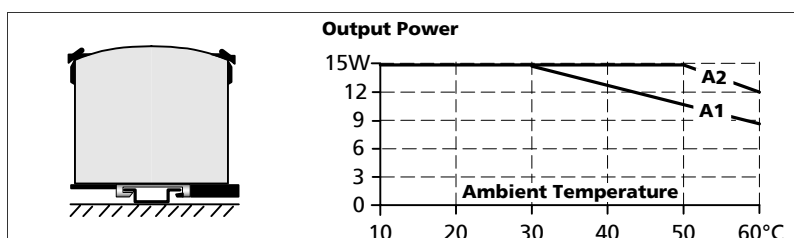


Fig. 25-11

Mounting Orientation D
(Horizontal cw)

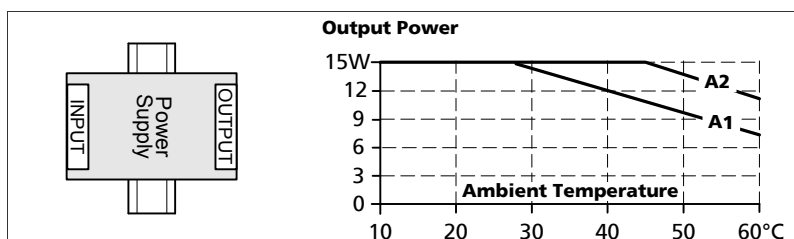


Fig. 25-12

Mounting Orientation E
(Horizontal ccw)

